

Advanced International Macroeconomics B (EI136)

Class 4

Dilemma-Trilemma, reserves and interventions

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Outline of today class

- Trilemma of policy in open economies.
 - Measures (Aizenman, Chinn and Ito 2022).
 - Empirical and theoretical assessments (Han and Wei 2017, Cesa-Bianchi, Ferrero, and Li 2025).
 - Limited use of exchange rate when there are not enough tools (Jeanne 2022).
- Foreign exchange reserves, and intervention (FXI).
 - Pattern, and currency composition (Arslanalp et al. 2024).
 - Theory on use of reserves and FXI (Céspedes and Chang 2024).
 - Foreign exchange interventions work: pattern and effect (Adler et al. 2021, Fratzscher et al. 2019).

TRILEMMA OR DILEMMA?

Trade-offs in international policy

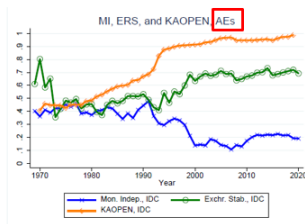
- Standard Trilemma of international macroeconomics: one can have only two out of three goals: a) independent monetary policy, b) stable exchange rate, c) capital mobility.
- Indices measuring the policy choice (Aizenman, Chinn and Ito 2022).
 - Exchange rate stability (*ERS*), inverse of variance of FX vis-à-vis a reference currency.
 - Monetary independence (*MI*), sensitivity of policy rate to rate in a reference country (US or Euro).
 - Capital account openness (*KAOPEN*), “Chinn-Ito” index of capital mobility.
- Evolution through time is contrasted across country groups.
 - Advanced: more KO, more ERS and less MI (driven by Euro).
 - Emerging: more KO, less ERS.
 - Developing: no major changes.

Indices of trilemma corners

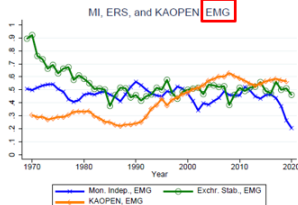
- Contrasted evolution across country groups.

Figure 2: Development of the Trilemma Configurations Over Time

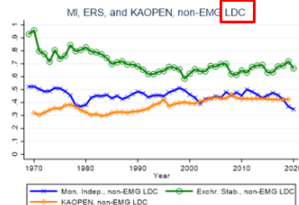
(a) Advanced Economies



(b) Emerging market economies



(c) Non-Emerging Market Developing Countries



Aizenman, Joshua, Menzie Chinn, and Hiro Ito (2022). "The Impacts of Crises on the Trilemma Configurations", NBER working paper 30406

Exchange rate regimes and policy autonomy

- Group countries according to FX regime (IMF classification) and capital mobility (Chinn-Ito index).
- Regression of changes in domestic policy interest rate on changes in anchor (US, euro area) policy interest rate. Autonomy is a small, or zero, coefficient.
 - Floating exchange rate raises autonomy (smaller coefficient on anchor), with gradual effect as a partial float already helps.
 - Capital controls raise autonomy, but need to be comprehensive (no effect of partial controls).
- Need to control for other (possibly common) drivers of the interest rates (Han and Wei 2017). Change in a country's rate reflects:
 - (Estimated) Taylor rule, because of changes in growth and inflation prospects (from WEO).
 - US rate, measuring the extent of policy autonomy (inverse).
 - Risk (VIX).
- The coefficient on the US rate should differ depending on the degree of capital mobility (Chinn-Ito index) and the exchange rate regime.

- Policy autonomy with capital controls, the interest rate does not follow the US rate.
- Without controls, a flexible exchange rate offers *some* autonomy.
 - Connection stronger after 1998, reflecting higher financial integration. Observed for short maturity and long maturity interest rates.
- Allow for a different connection to US interest rate depending on whether it increases or decreases.
 - No connection to rising US rates under a flexible exchange rate regime (let the domestic currency depreciate).
 - Connection to falling US rates (counter the appreciation of the domestic currency).
- Overall a “2.5-lemma” instead of a trilemma vs. dilemma.
 - Capital controls deliver autonomy. Flexible exchange rate delivers autonomy only when the US interest rate increases, but not when it decreases (fear of local currency appreciation).

- Autonomy (no sensitivity to US rate) when there are **capital controls**, regardless of FX regime.
- **Without capital controls**, more sensitivity to US rate under a fixed exchange rate than under a flexible one.

Table 3

Baseline model for both short-term policy rates and long-term government bond yields.

		Short-term 1990–2009 (1)	Short-term 1990–1998 ^a (2)	Short-term 1999–2009 ^b (3)	Long-term 1999–2009 (4)
$r_{L,t}^p - 1$	λ	-0.048 ^e (0.008)	-0.007 (0.015)	-0.11 ^e (0.01)	-0.071 ^e (0.013)
$\Delta GDP\ growth_{i,t}$	ϕ_1	0.096 (0.06)	0.237 (0.144)	0.044 (0.053)	0.027 (0.025)
$\Delta Inflation_{i,t}$	ϕ_2	0.329 ^e (0.048)	0.134 (0.096)	0.413 ^e (0.049)	0.172 ^e (0.039)
$D_{fixed,NC}\Delta r_{L,t}^{US}$	β_1	0.649 ^e (0.39)	0.402 (2.09)	0.648 ^e (0.297)	0.665 ^e (0.259)
$D_{flex,C}\Delta r_{L,t}^{US}$	β_2	0.034 (0.325)	1.998 (1.286)	-0.047 (0.262)	0.118 (0.302)
$D_{flex,NC}\Delta r_{L,t}^{US}$	β_3	0.450 ^e (0.176)	0.492 (0.438)	0.493 ^e (0.153)	0.473 ^e (0.111)
$D_{flex,C}\Delta r_{L,t}^{US}$	β_4	0.029 (0.127)	0.008 (0.334)	0.059 (0.109)	0.251 ^e (0.099)
ΔVIX_t	δ	0.23 (0.199)	0.086 (0.584)	0.172 (0.168)	0.177 ^e (0.089)
t-Test: $\beta_1 > \beta_3$		-0.473	0.042	-0.472	-0.682
t-Test: $\beta_4 < \beta_3$		1.998 ^e	-0.892	-2.416 ^e	-1.514
Adjusted R-squared		0.09	0.00 ^d	0.30	0.22
No. of Obs.		827	295	531	356

^a During 1990–1998, with available policy rate data, there is only one economy observed for the regime of fixed exchange rate and no capital controls (HKG).

^b Argentina 2001/April is an outlier and removed from the regression sample.

^c To test $\beta_1 > \beta_3$, we re-write the regression as $\Delta r_{L,t}^p = c + \lambda r_{L,t-1}^p + \phi_1 * \Delta GDP\ growth_{i,t} + \phi_2 * \Delta Inflation_{i,t} + \beta_1 (D_{fixed,NC}\Delta r_{L,t}^{US} + D_{flex,NC}\Delta r_{L,t}^{US}) + \beta_2 D_{flex,C}\Delta r_{L,t}^{US} + (\beta_3 - \beta_1) D_{flex,NC}\Delta r_{L,t}^{US} + \beta_4 D_{flex,C}\Delta r_{L,t}^{US} + \delta \Delta VIX_t + e_{i,t}$. In this form, the condition that $\beta_3 < \beta_1$ is the same as the coefficient, $(\beta_3 - \beta_1) < 0$.

^d The unadjusted R-squared for Column 2 is 0.027.

^e Significant at the 10%.

- Without capital controls, sensitivity to US rate under a **fixed exchange rate**.
- With a flexible rate, no sensitivity to a US rate **increase** (appreciates the dollar), sensitivity to US rate **decrease** (depreciates the dollar).

Table 4

Asymmetric responses under flexible exchange rate regimes plus an alternative Taylor rule specification.

		(1) Asymmetric responses	(2) Adding exchange rate	(3) Only countries with policy rates ^a	(4) Long-term bond yield as dependent variable
$r_{i,t}^p - 1$	λ	-0.114* (0.03)	-0.117* (0.03)	-0.116* (0.03)	-0.069* (0.04)
$\Delta GDP_{growth_{i,t}}$	ϕ_1	0.04 (0.07)	0.064 (0.08)	0.073 (0.08)	0.035 (0.03)
$\Delta Inflation_{i,t}$	ϕ_2	0.413* (0.09)	2.442* (1.36)	2.56* (1.33)	0.861 (0.57)
$\Delta Real\ Exchange\ Rate_{i,t}$	ϕ_3	- (0.17)	2.03 (1.34)	2.145 (1.31)	0.693 (0.57)
$D_{flex,NC} \Delta r_{i,t}^{IS}$	β_1	0.656* (0.17)	0.617* (0.17)	0.595* (0.17)	0.663* (0.19)
$D_{flex,C} \Delta r_{i,t}^{IS}$	β_2	-0.048 (0.15)	-0.039 (0.15)	-0.095 (0.14)	0.161 (0.14)
$D_{flex,NC} Pos\ \Delta r_{i,t}^{IS}$	$\beta_{3,pos}$	0.152 (0.23)	0.114 (0.24)	0.127 (0.25)	0.329* (0.12)
$D_{flex,NC} Neg\ \Delta r_{i,t}^{IS}$	$\beta_{3,neg}$	0.630* (0.160)	0.616* (0.170)	0.592* (0.17)	0.526* (0.12)
$D_{flex,C} Pos\ \Delta r_{i,t}^{IS}$	$\beta_{4,pos}$	0.215 (0.25)	0.192 (0.24)	0.169 (0.24)	0.001 (0.24)
$D_{flex,C} Neg\ \Delta r_{i,t}^{IS}$	$\beta_{4,neg}$	0.013 (0.16)	0.025 (0.16)	0.037 (0.16)	0.367* (0.22)
ΔVIX_t	δ	0.172 (0.15)	0.047 (0.18)	-0.032 (0.19)	0.116 (0.10)
Adjusted R-squared		0.304	0.307	0.303	0.223
No. of Obs.		531	531	514	356

Note: Robust standard errors clustered at the level of country-period are reported for all columns.

Bold and an asterisk means significant at the 10% level.

^a Argentina is excluded as it is an obvious outlier.

Han, Xuehui, and Shang-Jin Wei (2017) "International transmissions of monetary shocks: Between a trilemma and a dilemma" *Journal of International Economics* 110, pp. 205-219.

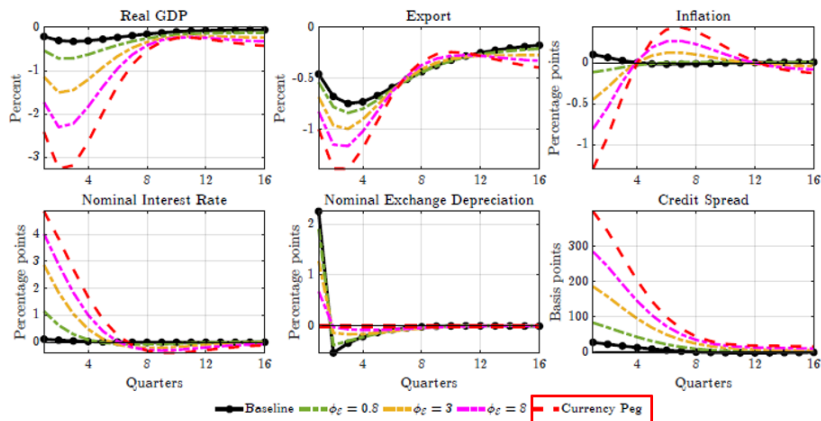
Exchange rate regime in a model

- Two country model with a financial accelerator channel (dollar debt) and the expenditure switching channel (Cesa-Bianchi, Ferrero, and Li 2025).
- Empirical evidence with multi-country VAR.
 - Tightening of US monetary policy is contractionary, in U.S. and rest of the world.
 - Contraction more limited for countries with flexible exchange rates.
- Two country model with domestic and financial frictions (similar to Akinci and Queralto 2024).
- Tightening of foreign (dominant currency) monetary policy raises spreads and lowers GDP.
 - A peg does not insulate, but moderates the impact.
 - Countercyclical taxes on domestic credit and / or foreign currency also limits volatility, especially under a float.

Impact of foreign policy tightening

- Recession and increase in spreads. Much less so (but not zero) under a floating exchange rate.

Figure 7: Comparing different degrees of exchange rate flexibility.



Cesa-Bianchi, Ambrogio, Andrea Ferrero, and Shangshang Li (2025). "Capital Flows and Exchange Rates: A Quantitative Assessment of the Dilemma Hypothesis", draft

Using other tools

- Macroprudential policy (tax on credit) and capital controls (tax on foreign borrowing) help.

Table 3: Exchange rate regimes, policy rules and macroeconomic volatility.

Regime (additional instrument)	Policy rules coefficients		Standard deviations	
	ϕ_{ε}	ϕ_k or ϕ_b	Real GDP	Inflation
Fully flexible exchange rate	0	0	1.22	0.07
Baseline	0.062	0	1.58	0.05
Baseline + tax on domestic credit (τ_k^{flex})	0.062	-0.05	0.95	0.07
Baseline + tax on foreign borrowing (τ_b^{flex})	0.062	-0.15	0.60	0.32
Peg	∞	0	109.66	8.79
Peg + tax on domestic credit (τ_k^{peg})	∞	-0.05	21.88	5.66
Peg + tax on foreign borrowing (τ_b^{peg})	∞	-0.35	22.46	4.77

Cesa-Bianchi, Ambrogio, Andrea Ferrero, and Shangshang Li (2025). "Capital Flows and Exchange Rates: A Quantitative Assessment of the Dilemma Hypothesis", draft

POLICY WITH UIP SHOCKS

Policy when tools are limited

- Why don't policy makers (especially in EM) make full use of exchange rate flexibility (Jeanne 2022)?
- Exchange rate movements reflect deviations from UIP:

$$(1 + i_t) \frac{E_t}{E_{t+1}} = (1 + i_t^*) (1 + \rho_t)$$

- Lower ρ_t implies a shift of investors towards local currency bonds (lower i_t).
- Simple small open economy model.
 - Endogenous production of traded and non-traded goods.
 - Wage stickiness in short run (1 period), then full flexibility.
 - Short run shock to UIP premium $\rho - \bar{\rho}$.
 - Inflows in local currency bonds (holdings B), outflows in dollar bonds (holdings B^*).

- Lower premium ($\rho - \bar{\rho} < 0$) leads to trade deficit ($x < 0$) and higher gross inflows and outflows:

$$x = b^* - b = \frac{1}{\sigma_T} (\rho - \bar{\rho})$$

$$b^* = -\frac{\phi}{\sigma_T} (\rho - \bar{\rho})$$

$$b = -\frac{1+\phi}{\sigma_T} (\rho - \bar{\rho})$$

- Lower cost of borrowing, so increase \$ investment and borrow through a trade deficit.

Policy response

- Trade-off between output gap in traded sector (reflects exchange rate e) and non-traded sector (reflects interest rate i):

$$y_T = \frac{e}{\gamma} \quad ; \quad y_N = -\frac{i - \bar{i}}{\sigma_N}$$

- UIP ($i - \bar{i} + e = \rho - \bar{\rho}$) implies that we cannot set $i - \bar{i} = e = 0$.
- Optimal (limited) policy: some appreciation of the real exchange rate, some reduction in interest rate (less than foreign rate: $i - \bar{i} < \rho - \bar{\rho}$).
- Planner's optimum: lower trade deficit x and smaller capital inflows b , no change to outflows b^* .
- Can be implemented through a tax on holding domestic bonds for all investors. They get $(1 + i_t) / (1 + \tau) < 1 + \bar{i}$.
 - Higher tax when UIP shock is low:

$$\tau = -(1 - \chi)(\rho - \bar{\rho})$$

- Tax inflows and outflows. Taxing only inflows is very suboptimal.
- Using tax and exchange rate flexibility leads to better welfare.

Welfare impact

- Tax on investment has a strong effect than FX flexibility.

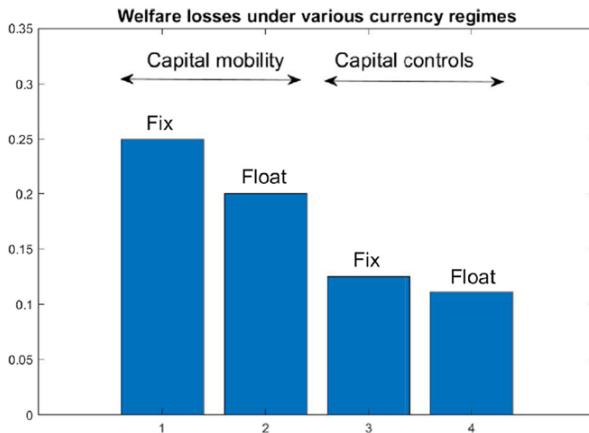


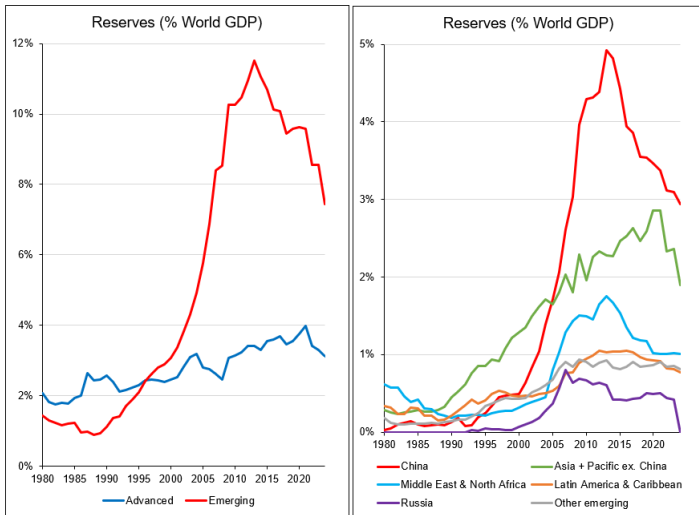
Fig. 2. Welfare losses under different policy regimes.

Jeanne, Olivier (2022). "Rounding the corners of the trilemma: A simple framework", *Journal of International Money and Finance* 122.

FOREIGN RESERVES

Rise in reserve accumulation

- Large accumulation of reserves by emerging economies until early 2010's, subsequent decrease.



- Several reasons to accumulate reserves.
 - Stabilize the exchange rate, or maintain a peg.
 - Ensure the ability to finance imports.
 - Ensure the ability to support the financial sector (e.g. when banks have foreign currency liabilities).
 - Mercantilism: hold the exchange rate undervalued to promote exports.
- Composition of reserves has evolved in recent years (Arsanalp et al 2024).
 - Growing share of gold in emerging economies.
 - Declining share of dollar.
 - Growing share of non-traditional reserve currencies (Oceania, small advanced economies).
 - Little recent progress of renminbi.

- Rising holdings of gold (and additional price effect).
- Steady decline in dollar share.

Figure 4. Gold Held by the Official Sector, 1950-2024

(million fine troy ounces)

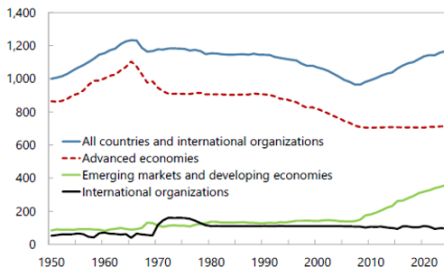
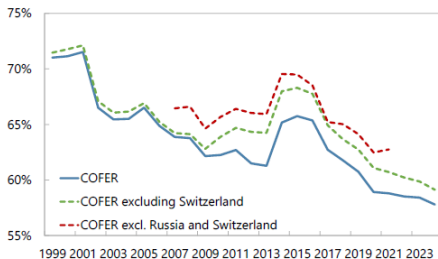


Figure 8. U.S. Dollar Share of Global FX Reserves, with and without Russia and Switzerland, 1999-2024

(Percent)



Arslanalp, Serkan, Barry Eichengreen, and Chima Simpson-Bell (2025). "Our Underappreciated International Reserve System", NBER working paper 34478

Drivers of the currency composition

- Panel regressions of macroeconomic and institutional factors.
- Gold share not much affected by macroeconomic conditions.
 - Higher share with a float (less need to hold foreign currencies) and independence.
- Currency share reflects macroeconomic factors and geopolitics.
- Higher share of dollar with strong US connections.
 - In economic terms (pegs, trade and financial exposure).
 - In political terms (military alliance).
- Renminbi share appears more political.
 - Higher with PBoC swap lines, Belt and Road Initiative.
 - Lower when there is an alliance with the US.
 - Lower with an independent central bank (no political pressure to hold renminbi).

- Higher share of gold with a flexible exchange rate and an independent central bank.

Table 6. Emerging Markets and Developing Economies: Gold Share of Reserves

Variable	(1)	(2)	(3)	(4)	(5)
Gold share (%)					
CPI inflation	0.17***	0.03	0.04	0.06	0.04
Fiscal balance	-0.05	0.11	0.09	-0.12	-0.18*
Public debt	0.06***	0.00	0.00	0.07***	0.07***
Real GDP growth	0.05	0.20	0.23*	0.08	0.06
Gold production	0.00***	0.00	0.00	0.00***	0.00***
Reserve assets growth	0.00	-0.01	-0.01	0.00	0.00
Currency appreciation	-0.01	-0.12*	-0.13*	-0.04	-0.04
Floating exchange rate	0.08***	0.03**	0.03**	0.05***	0.06***
Trade openness	0.01	0.00	0.01	0.01	0.01
US financial sanction		0.00	0.00	0.03***	0.03***
US defense pact			0.01*	0.02***	0.02***
Central bank independence				0.07***	0.06***
Financial development					-0.08***
Constant	-0.01	0.02	0.01	-0.07***	-0.05**
var(e)	0.01***	0.01***	0.01***	0.00***	0.00***
Statistics					
N	1705	789	789	604	604
AIC	-1388.81	-832.00	-832.56	-1196.31	-1207.28
BIC	-1225.57	-687.21	-683.09	-1050.99	-1057.56

Arslanalp, Serkan, Barry Eichengreen, and Chima Simpson-Bell (2025). "Our Underappreciated International Reserve System", NBER working paper 34478

Dollar share

- Higher with stronger connections with the U.S.: economic (peg, trade, finance), and political.

Table 7. US Dollar Share of Foreign Exchange Reserves

Variable	(1)	(2)	(3)	(4)	(5)	(6)
USD share (%)						
Peg USD	0.08***	0.08***	0.10***	0.13***	0.11***	0.10***
Peg EUR	-0.41***	-0.40***	-0.39***	-0.35***	-0.34***	-0.28***
Peg Other	-0.16***	-0.16***	-0.15***	-0.18***	-0.15***	-0.09
Trade USD	0.42***	0.45***	0.04	0.03	0.00	0.34**
Trade EUR	-0.26***	-0.25***	-0.22***	-0.18*	-0.12	-0.35***
Trade JPY	0.09	0.11	-0.28	0.17	0.42	0.79***
Trade GBP	-1.12**	-1.15**	-0.72	-1.40**	-1.06*	-0.46
Debt USD	0.06	0.06	-0.05	0.34***	0.41***	0.38***
Debt EUR	-0.01	-0.01	-0.17*	0.34***	0.39***	0.47***
Debt GBP	-3.65*	-3.65*	-3.30*	-8.25**	-8.11**	-9.24**
Debt JPY	-0.14	-0.13	-0.21*	0.03	0.00	0.05
EA dummy	0.28***	0.28***	0.27***	0.26***	0.29***	
US financial sanction		-0.02	-0.01	-0.01	-0.04	-0.04
US defense pact			0.13***	0.11***	0.15***	0.08**
Central bank independence				-0.23***	-0.26***	-0.24***
Financial development					-0.23***	-0.29***
Reserve assets growth						0.10*
Currency appreciation						-0.33**
Floating exchange rate						-0.05
Trade openness						-0.07
Constant	0.58***	0.58***	0.61***	0.31***	0.32***	0.40**
var(e)	0.04***	0.04***	0.03***	0.03***	0.03***	0.03***
Statistics						
N	676	676	676	507	507	386
AIC	-249.58	-248.05	-282.86	-197.14	-202.70	-213.13
BIC	-91.51	-85.47	-115.76	-28.00	-29.33	-43.03

Arslanalp, Serkan, Barry Eichengreen, and Chima Simpson-Bell (2025). "Our Underappreciated International Reserve System", NBER working paper 34478

Renminbi share

- Higher with trade, swap, and B&R connection (but not so robust).
- Lower when political connection with US and central bank independence.

Table 10. Chinese Renminbi Share of FX Reserves

Variable	(1)	(2)	(3)	(4)	(5)	(6)
CNY share (%)						
Peg USD	-0.10***	-0.10***	-0.12***	-0.10***	-0.01	-0.01
Peg EUR	-0.04***	-0.03***	-0.03***	-0.03***	-0.04	-0.05
Peg Other	-0.07***	-0.04**	-0.04**	-0.04**		
Trade USD	-0.11***	-0.08***	-0.08***	-0.09***	-0.32***	-0.31***
Trade EUR	0.13***	0.09***	0.09***	0.08***	-0.01	-0.01
Trade JPY	1.99***	1.54***	1.55***	1.65***	3.14***	3.11***
Trade GBP	0.48***	0.41***	0.43***	0.47***	0.51***	0.50***
Debt USD	-0.08***	-0.05	-0.06	-0.08*	0.83***	0.84***
Debt EUR	-0.11***	-0.09**	-0.10***	-0.11***	0.81***	0.82***
Debt JPY	0.01	-0.04	-0.07	0.13	3.06***	3.03***
Debt GBP	11.43***	10.43***	9.77***	10.69***	19.75***	19.72***
EA dummy	-0.10***	-0.09***	-0.09***	-0.09***	-0.01	-0.01
Trade CHN		0.02	0.02	0.08***	0.12	0.11
PBOC swap		0.01***	0.01*	0.01***	-0.01	-0.01
Belt and Road		0.02***	0.01***	0.01***	0.00	0.00
US financial sanction			0.01***	0.01***	0.00	0.00
Reserve growth				0.00	0.01	0.01
Currency appreciation				0.02***	0.00	0.00
Trade openness				0.08***	0.08***	0.08***
US defense pact					-0.04**	-0.04**
Central bank independence					-0.07***	-0.07*
Financial development						0.03
Constant	-0.02	-0.04	-0.03	-0.04	-0.93***	-0.95***
var(e)	0.00***	0.00***	0.00***	0.00***	0.00***	0.00***
Statistics						
N	556	519	519	517	187	187

Arslanalp, Serkan, Barry Eichengreen, and Chima Simpson-Bell (2025). "Our Underappreciated International Reserve System", NBER working paper 34478

RESERVES IN A MACRO MODEL

- Role of reserves in a model with occasionally binding constraints (Céspedes and Chang 2024).
- Households and firms in a small open economy borrow from domestic banks, who in turn borrow from foreign banks.
 - Borrowing constraints: banks' equity cannot be below a threshold that is proportional to its assets (i.e. lower bound on leverage).
 - This threshold is stochastic.
- If the constraint binds, or is likely to bind, banks deleverage and there is a credit crunch for investment and consumption.

- The central bank can purchase foreign assets, but these earn a relatively low return (liquidity premium τ).
- If the constraint binds, the central bank uses the reserves to recapitalize banks.
 - Sustains consumption and investment.
- Not optimal to accumulate enough reserves to rule out a crisis, as reserves are costly (τ).
- Reserves are more effective if used for banks, who leverage further, rather than directly for consumers and households.
- Reserves are more effective than initial tax on ex-ante borrowing, which amplifies distortion on investment.

Impact of reserves

- Lower risk of crisis when reserves F are high.
- It helps welfare, up to a point.

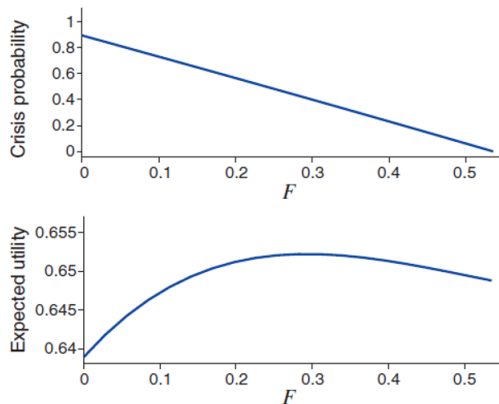


FIGURE 4. RESERVES, CRISIS PROBABILITY, AND EXPECTED UTILITY

Céspedes, Luis Felipe, and Roberto Chang (2024). "Optimal Foreign Reserves and Central Bank Policy under Financial Stress", *American Economic Journal: Macroeconomics* 16(3), pp. 230-267.

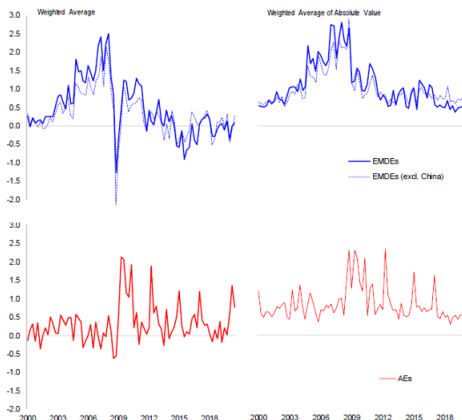
FOREIGN EXCHANGE INTERVENTIONS

- Sales and purchases of foreign reserves are a frequently used policy tool (Adler, Chang, Mano and Shao 2021).
 - Detailed database, with interventions on spot exchange rate markets as well as through derivatives.
 - Less so since the 2008 crisis.
- In emerging economies, as well as in several advanced ones (HK, Singapore, Switzerland).
- Interventions consist mostly of purchasing foreign currencies (counter an appreciation of one's own currency).

Historical evolution

- Reduction after the crisis in emerging economies.

Figure 5. Foreign Exchange Intervention by Group
(broad proxy, in percent of GDP)



Adler, Gustavo, Kyun Suk Chang, Rui Mano, and Yuting Shao (2021). "Foreign Exchange Intervention: A Dataset of Public Data and Proxies", IMF working paper 2021/047.

Does using reserves (FX intervention) work?

- Exchange rate intervention is quite common (Fratzscher, and al. 2019).
- Focus on sterilized intervention where $M0$ is unchanged (unsterilized intervention clearly works).
 - The central bank sells foreign currency assets and buys domestic currency assets.
- Theoretical channels for effectiveness.
 - Portfolio balance: private investors are willing to take the mirror position of the central bank only with different asset prices.
 - Signalling: the intervention conveys information about future policy.
- Challenge for empirical assessment: precise data on interventions are not common.
- Build an extensive database of daily observations for 33 countries from 1995 to 2011.
- Interventions are less frequent in floating regimes, and consist of foreign currency purchases most of the time.

How do we define success?

- An intervention can have different objectives.
- “Event” criterion: impact the level of the exchange rate.
 - Does the exchange rate move in the intended direction (depreciation of the domestic currency after a foreign currency purchase)?
- “Smoothing” criterion: limit volatility.
 - Is exchange rate volatility lower after the intervention than before (5 days windows)?
- “Stabilization” criterion: keep the exchange rate in a given range.
 - Does the exchange rate remain within a 2 percent band over the next two weeks?
- To assess effectiveness, test whether the criteria is met more often after an intervention than in other times.

Evidence of effectiveness

- Distinguish between exchange rate regime: floats, broad bands, narrow bands.
- Floaters are successful according to the event and smoothing criteria.
- Broad and narrow bands are successful according to the smoothing criterion.
 - Narrow bands are successful according to the stabilization criteria, but not to a large extent.
- Large interventions are more successful.
- “Oral” interventions (communications by the central banks) help, especially in turbulent markets.
 - Communication occurs in 52 % of the interventions.
- Effectiveness is stronger when comparing intervening countries with countries in similar situations but without intervention.

- Success of floaters in moving and smoothing the exchange rate.
- Success of band regimes in smoothing.

TABLE 4—UNCONDITIONAL SUCCESS RATES OF INTERVENTION EPISODES BY REGIME

Regime	Free floater		Broad band		Narrow band	
Criterion	Event (1)	Smooth (2)	Smooth (3)	Stabilize (4)	Smooth (5)	Stabilize (6)
Intervention episodes	0.611	0.883	0.791	0.348	0.781	0.840
Placebo rates	0.481	0.401	0.396	0.495	0.342	0.768
p -value (H_0 : equal effectiveness)	0.012	0.000	0.000	0.000	0.000	0.000
p -value (H_0 : actual $\leq$ placebo)	0.006	0.000	0.000	1.000	0.000	0.000
Actual events	95	77	561	1,062	1,010	2,893

Fratzcher, Marcel, Oliver Gloede, Lukas Menkhoff, Lucio Sarno, and Tobias Stohr (2019). "When is foreign exchange intervention effective? Evidence from 33 countries", *American Economic Journal: Macroeconomics* 11(1), pp. 132-156.